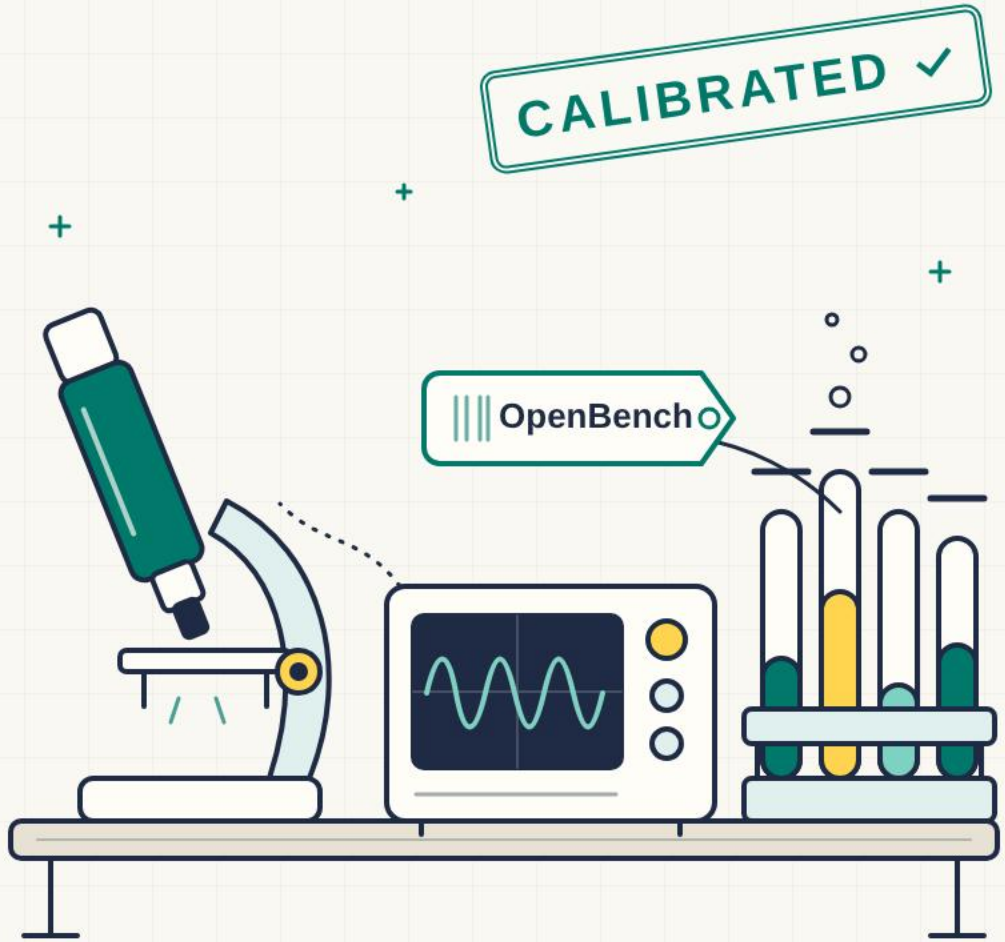
The Idle Instrument Paradox

From a booking portal to a network where a booked slot is a delivered result

Team Inferno · C. V. Raman Global University · IIT Bhubaneswar
Mahi · Minakshi Singha · Swapnil Behera

HYPOTHESIS

Booked should mean delivered.



Sell delivered results, not listings, and take ₹1.2 Cr in two tranches

WHAT OPENBENCH BECOMES

92%



on-time by Month 12 (78% today)

A managed "Assured" tier for trust-critical labs; cheaper tiers for price-sensitive makers

WHERE IT GROWS

17



host labs by Month 12 (ask: 12)

Pilot city, then deep-tech, startup hub, and manufacturing last

WHAT IT EARNS

M15



monthly break-even (ask: M18)

Uses ₹38.6L of ₹120L; ₹78.6L in the downside

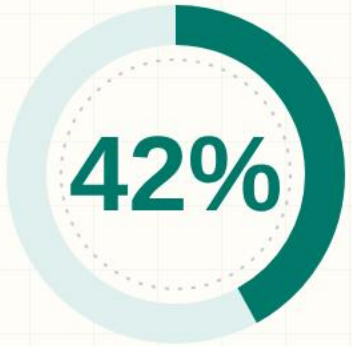
₹60L now → 🔒 **Month-7 gate** → **₹60L**

Investor: renegotiate, don't reject. ₹60L now + ₹60L at a Month-7 reliability gate. Keep 12 hosts, 90% on-time, M18 break-even; set 2,200 bookings and 55% repeat.

RESULT: Reliability, not discovery, is what customers will pay for, so we **fix it before we scale**.

Sources: case Exhibits 1–3; Team Inferno financial model (OpenBench_Financial_Model.xlsx, Base and Downside scenarios).

Customers leave because supply is listed, not productised: three structural causes



of customers return within 90 days

1. **Supply is listed, not productised**
   44% bookable; 14% cancelled, 8% late
2. **Labs earn too little per hassle**
   Flat 18% fee; labs absorb the admin
3. **No trust layer exists**
   31% cite turnaround; 27% custody

NOT JUST OPENBENCH



of requests on India's free I-STEM portal were executed (44,084 of 88,887).

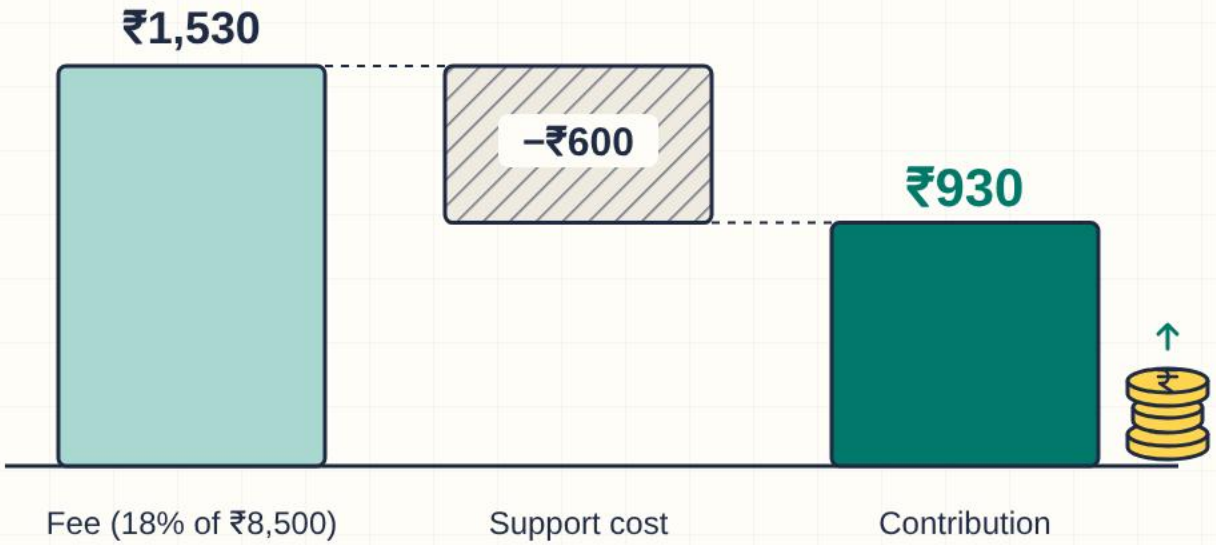
WHO LEAVES **Deep-tech & startups** fear IP leaks (27%) → Assured custody · **Manufacturers** need price + predictable turnaround (31%) → Connect + Priority SLA · **Seasonal users** miss slots → committed slots. Inference from Exhibits 1–2; detail in A8.

RESULT: A better listing won't fix this; **a guaranteed outcome will**. That is OpenBench's wedge.

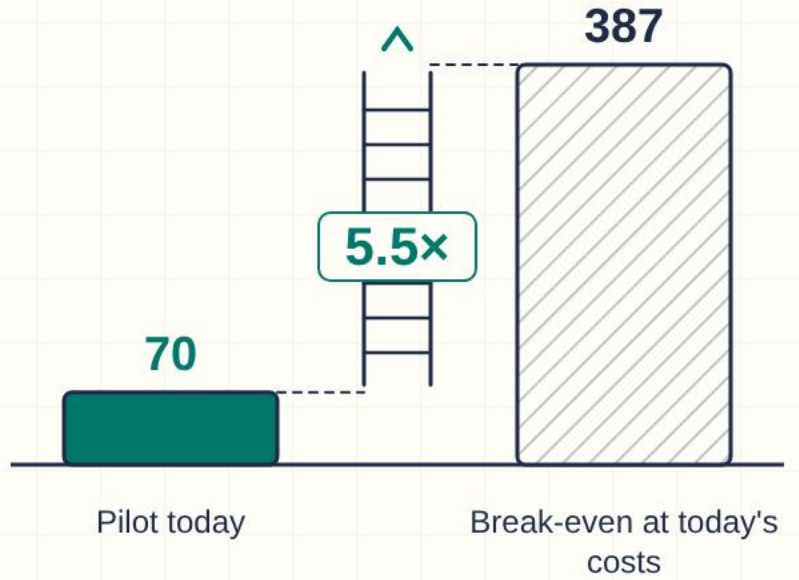
Sources: Exhibit 1 (pilot data, 60 interviews); [S1] Office of the PSA, I-STEM national footprint as on 31 Dec 2025.

At an 18% fee the pilot loses ₹2.95L monthly and needs 5.5× its volume

PER BOOKING (₹)



BOOKINGS PER MONTH



RESULT: **70 × ₹930 = ₹65,100 vs ₹3.6L fixed.** More volume on broken reliability multiplies cancellations.

Sources: Exhibit 1 (₹8,500 average booking, 18% fee, ₹600 support cost, ₹3.60L fixed, 420 bookings in 6 months); model tab 'Unit Economics'.

A segmented hybrid beats every pure model on reliability and investor fit

Criterion (weight)	Market-place	Managed	Lab SaaS	Our hybrid
Contribution per host (20)	3	4	2	4
Reliability (on-time) (25)	1	5	3	4
Lab adoption (15)	5	4	1	4
Onboarding capital (10)	5	2	3	3
Scalability (10)	5	2	4	4
Investor-KPI fit (20)	1	4	2	5
Weighted score /100	**56**	**77**	**48**	**82**

Shading: 1 (white) to 5 (darkest). Exhibit 3: ₹18.6k / ₹20.6k / ₹17.6k per host; on-time 82% / 94% / 86%; 4 of 10 labs would pay a subscription.

One backbone, three price points: scheduling, custody, escrow and reports are shared; only Assured runs until the M7 gate (Connect from M9).

Customer promise

Booked = delivered: committed slot, sealed custody, on time or 20% back.

Who pays

Customer pays; we keep 28% (Assured) or 8% + ₹12k lab fee (Connect).

Who operates

We run scheduling, custody, reports, disputes; the lab runs the instrument.

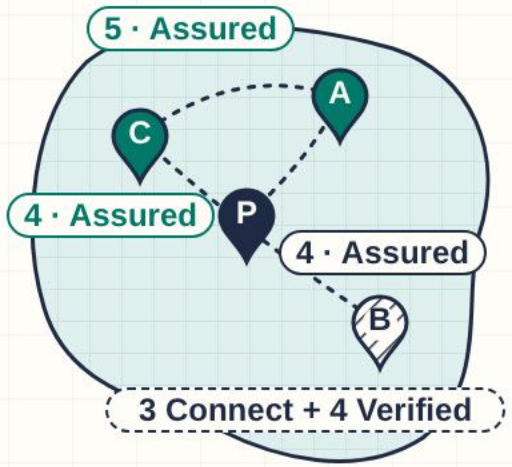
Why labs join

A pilot lab earns ₹1,28,775 vs ₹1,21,975 a month (+5.6%), with no admin.

RESULT: **Assured where trust is scarce, Connect where price is:** one network, two promises, priced to each cluster.

Sources: Exhibit 3; [S2] DST SRIMAN Guidelines 2022 (incentives for sharing); [S4] Science Exchange CEO on B2B take rates; team scoring.

Enter where trust pays most first; take the price-sensitive corridor last

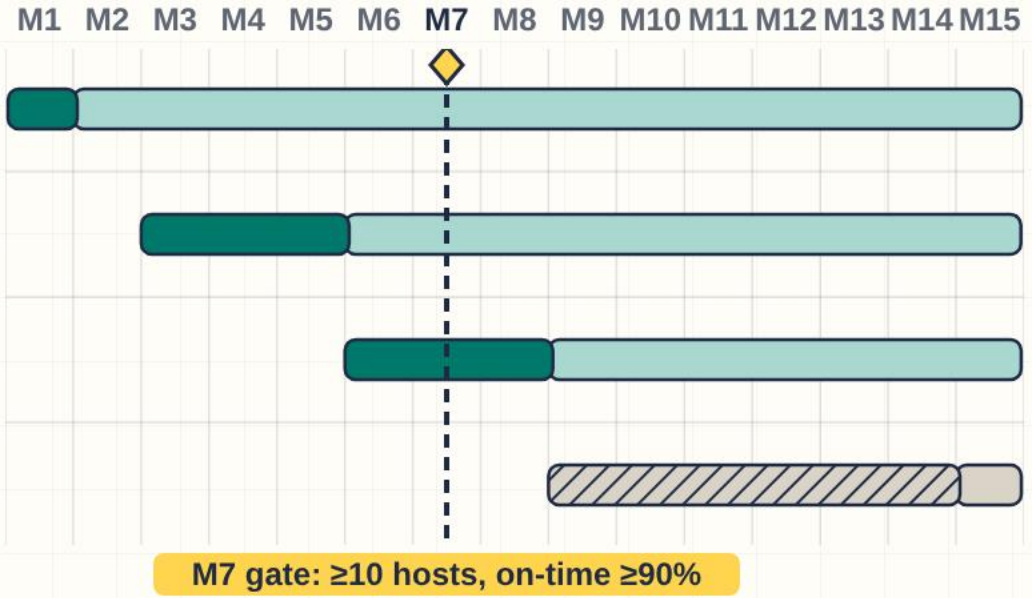


Pilot city 4 hosts · Assured
Convert existing labs, M1

A · Deep-tech 5 hosts · Assured
Go-live M3–M5

C · Startup hub 4 hosts · Assured
Go-live M6–M8

B · Manufacturing 3 Connect + 4 Verified
Go-live M9–M14



A first Highest value, ₹48,240/host/month; custody answers its IP worry.

C next Same startup customers as A; reuses the playbook. ₹23,140/host.

B last Most volume, thinnest margin; Connect wins below 26.7 bookings/host.

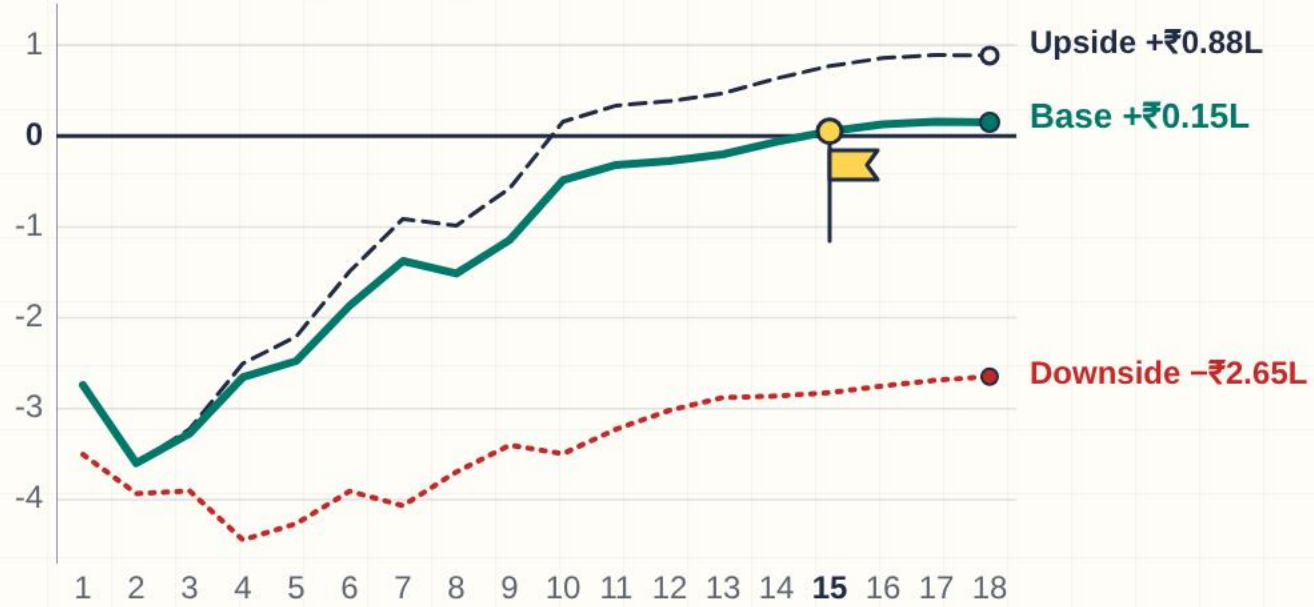
How we win customers incubator partners + I-STEM leads (80% free) · paid CAC ₹1,500 · paid back in ~1 booking · Funnel: A7

RESULT: **13 hosts in 3 cities by Month 8** meets the investor's footprint before we touch the riskiest cluster.

Sources: Exhibit 2 (projections, not guaranteed); A1 85% capture (pilot runs at 87.5% of baseline); A2–A5 ramp 30/60/85/100%; tab 'Unit Economics'.

Base case breaks even in Month 15; downside keeps ₹41.4L cash but misses M18

MONTHLY EBITDA (₹ LAKH) · MONTHS 1–18



M15
break-even, base

2,316
paid bookings M1–M12

₹38.6L
cash used, base (of ₹120L)

₹78.6L
cash used, downside

WHAT MOVES M18 EBITDA MOST

Driver	Low	High
Booking value ±10%	−₹0.84L	+₹1.14L
Capture 70% / 100%	−₹0.58L	+₹0.88L
Fixed cost +15/−10%	−₹0.67L	+₹0.70L

Downside + M9 response: loss narrows to −₹2.32L/mo; cash floor ₹47.7L (A9)

RESULT: Break-even rests on **two levers we will track monthly**; with both off, Month-18 EBITDA is −₹1.23L. Our +50% Priority price matches the express surcharge labs already publish [S9].

Downside: 65% capture, −10% booking value, +5% cancellations, 2-month delay, +15% fixed cost. Full tornado in appendix A5. Tab 'Scenarios'.

Accept the money, renegotiate two targets, and give the investor a tighter quality bar

Condition	Ask	Our base		Downside	Our stance
Hosts by M12	12 in 3 cities	17 in 4 cities	✓	15	Accept: 13 in 3 cities by M8
Paid bookings, M1–M12	2,400	2,316	×	1,533	Revise to 2,200
On-time at M12	90%	92.3%	✓	93.4%	Accept + cancellations ≤5%
90-day repeat at M12	60%	57.8%	×	58.0%	Revise to 55% (Assured 60%)
Operating break-even	by M18	M15	✓	after M18	Accept; triggers at M9
Capital	₹1.20 Cr	₹38.6L used	✓	₹78.6L used	₹60L now + ₹60L at M7

Why 55%: the case has one data point (78% on-time ↔ 42% repeat); published evidence implies 0.55–0.83 pt of retention per pt of on-time [S13], which gives 51–55% at M12 (A5). Bookings stay short of 2,400 with every cited fix, so we do not stretch the plan.

WHAT THE INVESTOR GETS



- ≤₹60L at risk until reliability is proven (downside uses ₹37.6L by M7)
- On-time ≥90% for 3 months; cancellations ≤5%
- Board observer and monthly KPI pack

RESULT: We promise only what our own model delivers, and **pay for the misses with a stricter quality bar**.

Sources: case funding terms; model tab 'Investor Test'. × = ask not met in our base case.

A six-step trust layer turns a listed instrument into a guaranteed result

1 

Slot

Only staffed slot blocks bookable

↑ Cancellations

2 

Sample

QR custody, sealed, photo-logged

↑ Confidentiality

3 

Test

Calibration checked; expiry delists

↑ Quality

4 

Report

Client owns data; 3-party NDA

↑ IP / trust

5 

Pay

Escrow; lab paid in 7 days

↑ Lab adoption

6 

Resolve

48h mediation, then re-test

↑ Repeat rate

	Listing portal	OpenBench Assured
Unit sold	Access to an instrument	A delivered, confidential result
Accountable	Nobody	OpenBench, under an SLA
Typical outcome	≈50% of requests executed (I-STEM)	94% on-time target (Exhibit 3)

When something breaks	Rule
Lab cancels	Backup lab; 10% payout held
Late result	20% of fee refunded
Priority SLA	+50% price; lab keeps half

RESULT: **Customers pay 28% for accountability;** the 86-versus-38 listing gap closes because only real slots are sold.

Sources: Exhibits 1 & 3; [S1] I-STEM execution data; [S3] I-STEM user guide; [S5] NABL / ISO/IEC 17025.

Four checkpoints and five KPIs show the board early whether it works



Month 3
6 hosts live
232 bookings
On-time 86%
Trust layer v1 live

Month 6
10 hosts live
669 bookings
On-time 94%
Repeat 50%

Month 12
17 hosts, 4 cities
2,316 bookings
On-time 92%
Repeat 58%

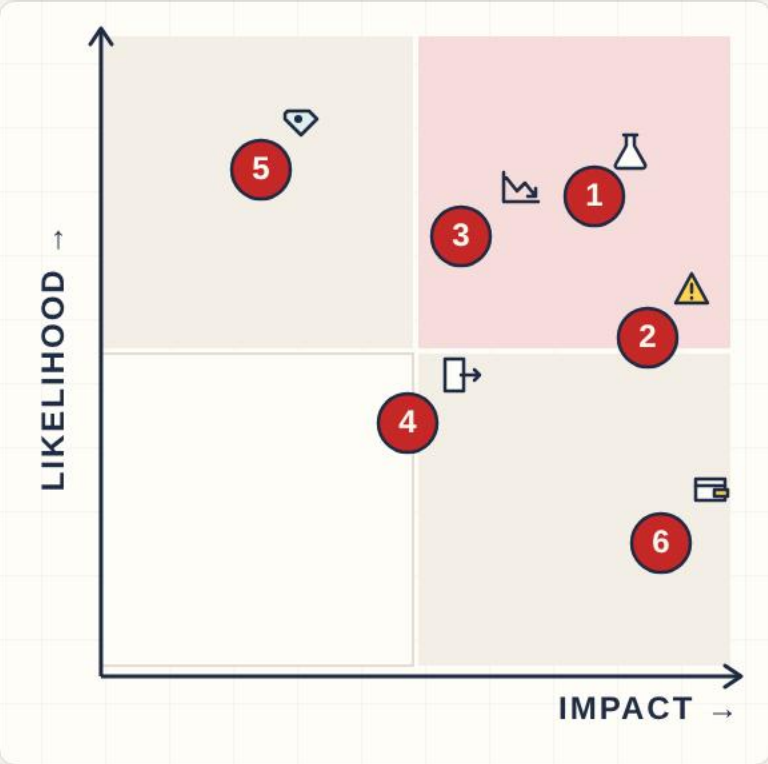
Month 18
20 hosts live
4,845 bookings
On-time 90%
EBITDA +₹0.15L/mo

KPI	Today		Target	today (grey) → target (teal tick)
On-time fulfilment	78%	→	**≥90% from M7**	
Host cancellations	14%	→	**≤5% by M6**	
Bookable share of listings	44%	→	**≥80%**	
90-day repeat	42%	→	**≥55% by M12**	
Contribution per booking	₹930	→	**≥₹1,250 (model ₹1,354)**	

RESULT: **On-time and cancellations decide the Month-7 tranche gate,** so the board watches those two first.

Sources: Exhibit 1 (today); model tab 'Monthly_Base' (milestones, base case).

Every major risk has a numeric trigger and a pre-agreed response



#	Risk	Trigger	Response
1	**Labs won't commit slots**	<10 hosts live at M7	Hold tranche 2; pitch SRIMAN credit
2	**Reliability slips**	On-time <88% at M6	Freeze new cities
3	**Demand below plan**	<70% of Exhibit 2 at M9	Skip B Verified; cut ₹0.5L/mo
4	**Customers go direct**	Repeat share of GMV falls 2 months	Escrow, guarantee, repeat pricing
5	**Price pushback in B**	B on-time <85% or churn >10%	Move host to Connect, or exit
6	**Cash runs low**	Cash <₹40L before M12	Stop onboarding

RESULT: Approve the Assured-first plan and the two-tranche offer. *Booked should mean delivered.*

Sources: [S6] Stanford Daily on core facilities declining to list; [S7] platform-leakage analysis; model downside scenario.

A1 · 18-month P&L, base case (₹ lakh unless noted)

	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15	M16	M17	M18
Active hosts	4	4	6	8	9	10	12	13	14	15	16	17	18	20	20	20	20	20
Bookings	70	73	89	117	145	174	205	234	263	290	315	341	366	399	425	441	449	449
Revenue	1.89	1.98	2.54	3.56	4.65	5.67	6.60	7.36	8.06	8.57	8.88	9.06	9.30	9.66	9.96	10.15	10.24	10.24
Variable cost	0.94	0.99	1.20	1.58	1.96	2.35	2.77	3.16	3.47	3.30	3.42	3.54	3.67	3.86	4.02	4.12	4.16	4.16
Contribution	0.94	0.99	1.34	1.98	2.68	3.32	3.83	4.21	4.59	5.27	5.46	5.52	5.62	5.80	5.94	6.04	6.08	6.08
Fixed cost	3.60	4.50	4.50	4.50	5.00	5.00	5.00	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50
Acquisition	0.09	0.09	0.11	0.14	0.16	0.18	0.20	0.22	0.23	0.26	0.28	0.30	0.33	0.36	0.39	0.41	0.42	0.43
EBITDA	-2.74	-3.60	-3.27	-2.65	-2.48	-1.87	-1.38	-1.51	-1.15	-0.49	-0.32	-0.28	-0.20	-0.07	**0.05**	**0.13**	**0.15**	**0.15**
Onboarding	2.40	2.40	2.40	1.20	1.20	2.40	1.20	0.70	0.70	0.70	0.45	0.45	0.90	0.00	0.00	0.00	0.00	0.00
Cash balance	114.9	108.9	103.2	99.3	95.7	91.4	88.8	86.6	84.8	83.6	82.8	82.1	81.0	80.9	81.0	81.1	81.2	81.4
On-time	81%	83%	86%	89%	91%	94%	94%	94%	94%	93%	93%	92%	92%	91%	90%	90%	90%	90%

Source: OpenBench_Financial_Model.xlsx, tab 'Monthly_Base'. Revenue includes the Priority-SLA share and Connect subscriptions. EBITDA = contribution − fixed − acquisition (A31–A33).

A2 · Assumption register (team assumptions labelled; case data never overridden)

ID	Assumption	Base	Downside	Basis
A1	Capture of Exhibit 2 mature volume	85%	65%	Pilot runs at 87.5% of the 20/host baseline
A2–5	New-host ramp (months 1/2/3/4+)	30/60/85/100%	same	Team assumption
A7	Pilot hosts after Assured conversion	20/month	same	Exhibit 3 baseline
A9–10	Booking value; extra cancellations	0%; 0%	−10%; 5%	Downside stress
A11	Onboarding delay for new hosts	0	+2 months	Institutional approvals [S6]
A12–16	Fixed: ₹3.6L + ₹0.4L trust lead + ₹0.5L per new city	₹5.5L max	+15%	Exhibit 1 + team assumption
A20	Assured variable cost from M10	₹1,200	same	11% learning: batching, custody kits
A22–24	Priority SLA: 15% uptake, +50% price, OpenBench keeps half	on	same	31% cite turnaround (Ex.1); 50% express surcharge [S9]
A25	Trust layer reaches tier on-time	6 months	same	Team assumption
A27	Repeat rises 1pt per 1pt on-time, 1-quarter lag	1.0×	same	Pilot: 78% on-time ↔ 42% repeat; 0.5–1.5× tested (A5) [S13]
A31	CAC per paid-acquired customer org	₹1,500	same	Events/webinars; avoids ₹50k+ B2B CAC [S10]
A32	New customer orgs per booking	0.405	same	Exhibit 1: 170 orgs / 420 bookings
A33	Customers won through free channels	80%	same	Incubators [S11], SATHI [S12], I-STEM leads, referrals

Source: OpenBench_Financial_Model.xlsx, tab 'Assumptions' (36 inputs, all editable; every output recalculates; A34–A36 drive the funnel and the downside response).

A3 · Unit economics by model and cluster

Model at Exhibit 3 baseline (₹8,500, 20/host)	Revenue/booking	Variable cost	Contribution/booking	Contribution/host/month	Onboarding	Payback (months)
B. Managed fulfilment (Assured)	₹2,380	₹1,350	₹1,030	**₹20,600**	₹1,20,000	5.8
C. Lab SaaS + booking fee (Connect)	₹680	₹400	₹280	**₹17,600**	₹70,000	4.0
A. Marketplace (Verified)	₹1,530	₹600	₹930	**₹18,600**	₹45,000	2.4

Cluster (mature Exhibit 2 volume)	Booking value	Bookings/host	Verified (18%)	Assured (28%)	Connect (₹12k + 8%)
Cluster A	₹12,000	24	₹37,440	₹48,240	₹25,440
Cluster B	₹6,500	30	₹17,100	₹14,100	₹15,600
Cluster C	₹8,000	26	₹21,840	₹23,140	₹18,240

🔒 Contribution per host per month. In Cluster B, Connect beats Verified whenever a lab does fewer than 26.7 bookings a month (89% of Exhibit 2).

Source: OpenBench_Financial_Model.xlsx, tab 'Unit Economics' (formulas from Exhibits 2 and 3).

A4 · SLA, penalty and confidentiality terms (outline, not legal advice)

Event	Customer gets	Lab bears	OpenBench does
Host cancels a committed slot	Rerouted job, same price	10% of payout held back	Reroutes to backup lab within 24h
Result late vs SLA	20% of fee refunded	Lower reliability score	Pays rebate from its fee
Customer cancels <48h	50% of fee charged	Paid for the reserved slot	Releases slot to waitlist
Disputed result	Free neutral re-test if lab at fault	Re-test cost if at fault	Mediates within 48h
Calibration expired	Instrument hidden from search	Must upload new certificate	Auto-delists on expiry

Standard 3-party agreement

- Customer owns samples, raw data and reports
- No copies kept beyond the retention window
- Anonymised sample IDs
- No reuse without written consent
- Liability capped at booking value, insured

Team design; for a real deployment these terms would need review by a lawyer.

A5 · Sensitivity of Month-18 EBITDA (₹ lakh change vs base)



A27 on-time → repeat (evidence 0.55–0.83× [S13])	0.5×	1.0× (base)	1.5×
90-day repeat at M12 (investor ask 60%; our offer 55%)	49.9%	**57.8%**	65.7%
90-day repeat at M18	48.2%	**54.4%**	60.5%

Base M18 EBITDA +₹0.15L. Both levers off (no Priority SLA, no cost learning): −₹1.23L. Source: tabs 'Sensitivity', 'Repeat check'.

A6 · Research sources (accessed October 2026)

[S1] Office of the PSA, I-STEM national footprint (31 Dec 2025): 88,887 requests generated, 44,084 executed — psa.gov.in/CMS/web/i-stem

[S2] DST, SRIMAN Guidelines 2022: sharing promoted through incentives and rating systems — indiascienceandtechnology.gov.in

[S3] I-STEM Quick Reference Guide for Users: users and custodians settle payments and reports directly — istem.gov.in

[S4] Bowery Capital interview with Science Exchange's CEO: too-high B2B take rates become untenable — bowerycap.com

[S5] NABL: government-authorised accreditation body for testing and calibration labs under ISO/IEC 17025

[S6] Stanford Daily: School of Medicine core facilities declined to list on Science Exchange — stanforddaily.com

[S7] Platform Chronicles, 'Platform leakage': leakage worsens as transaction value rises — platformchronicles.substack.com

[S8] DST, SATHI centres: shared, professionally managed facilities open to startups and MSMEs — dst.gov.in

[S9] Auriga Research, testing charges: 50% surcharge for express testing — aurigaresearch.com/testing-charges

[S10] upGrowth, CAC benchmarks for Indian B2B SaaS (2026): partnership CAC ₹50k–1.5L — upgrowth.in

[S11] Atal Innovation Mission: 72 Atal Incubation Centres — en.wikipedia.org/wiki/Atal_Innovation_Mission

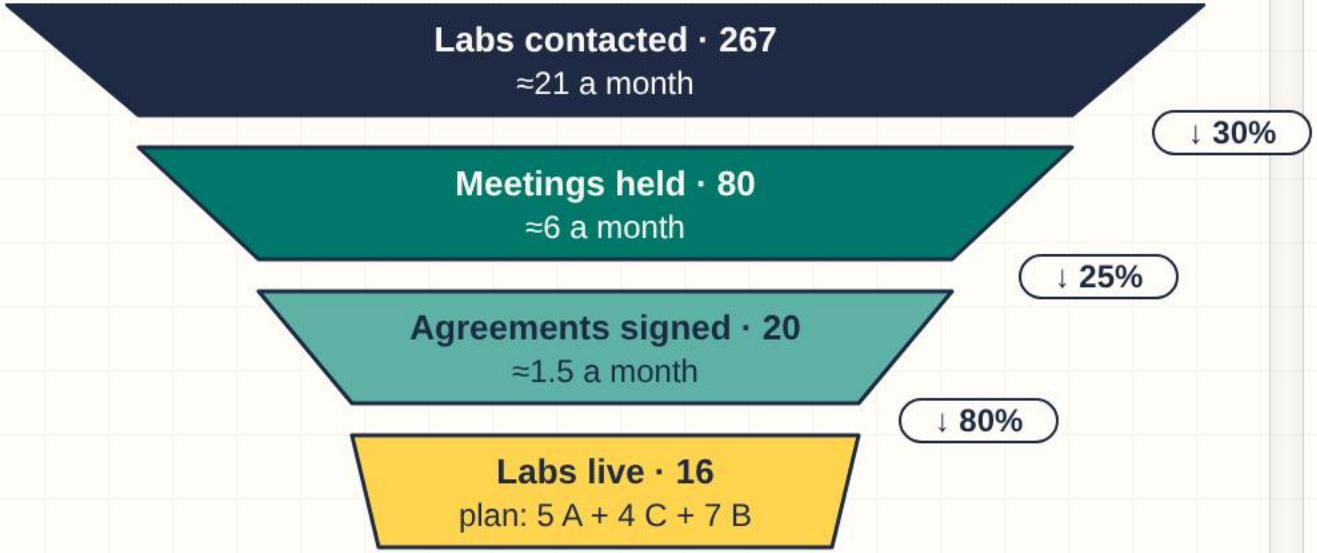
[S12] IMPRI, SATHI 2022-23: 5 centres, 32,000+ users, ~25% MSMEs/startups — impriindia.com

[S13] MMH / Grant Thornton: retention 52% at ≤50% on-time vs 76% at 79%; 82% at 90% — mmh.com

All case figures come from the Tech Zephyr case exhibits; external facts are context only, never a replacement for case data.

A7 · Lab onboarding funnel: 267 labs contacted to put 16 new labs live

LAB ONBOARDING FUNNEL · SPREAD OVER 13 MONTHS



Overall 6% of contacted labs go live. Contact → meeting 30% (A34); meeting → signed 25%; signed → live 80% (team assumptions).

WHERE THE 267 LEADS COME FROM

- **I-STEM** 3,812 institutes list 30,249 instruments [S1]
- **SATHI** 5 shared centres, 32,000+ users, ~25% MSMEs and startups [S12]
- **AIC** 72 Atal Incubation Centres: anchor labs and customers [S11]
- **Pilot** 4 labs and 170 customer organisations already on OpenBench

Slow-onboarding check	Break-even	M18 EBITDA	Lowest cash
Plan: 16 new labs (20 live by M14)	M15	+₹0.15L	₹80.9L
Slow: only 12 new labs (16 live)	Not by M18	−₹0.32L	₹81.0L

Source: model tab 'Funnel' (same engine; slow case drops the last four Cluster B labs). Lead pools: [S1] psa.gov.in I-STEM; [S11]; [S12]. Cash never drops below ₹80L in either case.

A8 · Who leaves and why, by segment (inference from Exhibits 1–2)

INFERENCE The case gives no repeat rate by segment, so we do not invent one. This maps the reasons customers gave (Exhibit 1) to the clusters' key issues (Exhibit 2).

Segment	Why they leave	Case evidence	What fixes it	Tier
Deep-tech labs and spin-outs (Cluster A)	Novel samples and data could leak	27% cite confidentiality (Ex.1); A key issue: IP protection, custom protocols (Ex.2)	Sealed custody, NDA, named analyst	**Assured**
Startups (Cluster C)	Slots vanish in busy seasons; small batches get bumped	31% cite turnaround (Ex.1); C key issue: seasonal demand, small batches (Ex.2)	Committed slots; Priority SLA for rush work	**Assured**
Manufacturers / MSMEs (Cluster B)	Price, and turnaround they cannot plan around	B key issue: price sensitivity, predictable turnaround (Ex.2); 31% turnaround (Ex.1)	Lower fee, fixed weekly slots, Priority on demand	**Connect / Verified**
All pilot customers today	Bookings fall through	14% cancelled, 8% late; 42% return in 90 days (Ex.1)	On time or 20% back; backup lab	**Assured**

What we track: 90-day repeat by cluster from Month 1, so this inference becomes measured data by the M7 gate.

Sources: Exhibit 1 (pilot data, 60 interviews: 31% turnaround, 27% confidentiality); Exhibit 2 (cluster key issues). Referenced from slide 3.

A9 · Four scenarios, including the downside with our M9 response fired

Output	Base	Downside	Upside	Downside + response
Break-even month	M15	Not by M18	M10	**Not by M18**
Hosts live at M12	17	15	17	**15**
Paid bookings M1–M12	2,316	1,533	2,559	**1,533**
On-time at M12	92.3%	93.4%	92.2%	**93.4%**
90-day repeat at M12	57.8%	58.0%	57.8%	**58.0%**
EBITDA at M18 (per month)	+₹0.15L	−₹2.65L	+₹0.88L	**−₹2.32L**
Cash used M1–M18	₹38.6L	₹78.6L	₹30.0L	**₹72.3L**
Cash used by the M7 gate	₹31.2L	₹37.6L	₹29.9L	**₹37.6L**
Lowest cash balance	₹80.9L	₹41.4L	₹86.0L	**₹47.7L**
Customer-acquisition spend M1–M18	₹4.6L	₹3.1L	₹5.2L	**₹2.9L**

RESPONSE FIRED AT M9 (SLIDE 11)

Skip the four Cluster B Verified labs (A35) and cut the City-4 cost of ₹0.5L a month (A36). The monthly loss narrows by ₹0.33L and the cash floor rises by ₹6.3L.

WHAT IT DOES NOT FIX

Break-even still slips past M18. "Stop onboarding below ₹40L cash" never fires: the floor stays at ₹41.4L even unmanaged. The response limits the loss; it does not rescue M18.

Source: model tabs 'Scenarios' and 'Monthly_DownResponse'. Downside: 65% capture, −10% booking value, +5% cancellations, 2-month delay, +15% fixed cost.